Excellent

第七周

P.118 A组

(4) $y=-2x^2+8x-5$

$a=-2 \Rightarrow a<0 \Rightarrow \cap$

顶点: $x=\frac{-b}{2a}=\frac{-(8)}{2(-2)}=\frac{-8}{-4}=2$

$y=-2(2)^2+8(2)-5$

$y=-2(4)+16-5$

$y=-8+16-5$

$y=3 \Rightarrow (2,3)$

答: 在 $(-\infty,2)$ $y=-2x^2+8x-5$ 是单调递增

而在 $(2,+\infty)$ 是单调递减。值域是 $(-\infty,3]$

(5) $y=x^2-2x-3$

纵轴: $x=0$ 时, $y=-3 \Rightarrow (0,-3)$

横轴: x -3 / x 1

$(x-3)(x+1)=0$

$x=3,-1$

$\Rightarrow (3,0)$ 和 $(-1,0)$

(7) $y=x^5$ 定义域, 值域, 奇偶, 单调性

定义域: $\mathbb{R}$

值域: $\mathbb{R}$

奇函数

单调递增

(8) $y=x^{-2}$ 定义域, 值域, 奇偶性

$y=x^{-2}=\frac{1}{x^2}$

定义域: $(-\infty,0)\cup(0,+\infty)$

值域: $(0,+\infty)$

偶函数

(9) $y=\frac{k}{x}$ 过 $(-3,4)$

$4=\frac{k}{-3}$ 解析式: $y=\frac{-12}{x}$

$-12=k$

(11) $y=\frac{x^2-9}{x-3}$ $x\neq 3$

$y=\frac{(x-3)(x+3)}{(x-3)}$

$y=x+3,(x\neq 3)$

$x=3$

定义域: $\mathbb{R}\setminus\{3\}$

因为原式 $\frac{x^2-9}{x-3}$ 分母不能等于0, 原式的限定就是 $x\neq 3$, 所以 $x+3$ 在 $x=3$ 是空点。

(14) $y=|x|$ ↑1, ←2

$y=|x|$

$x=-2$

$y=1+|x+2|$

P.119 B组

(3) $y=kx+2k+7$ 与 $y=x^2-2x$ 有一个交

$\begin{cases} y=kx+2k+7 \\ y=x^2-2x \end{cases}$

$kx+2k+7=x^2-2x$

$0=x^2-2x-kx-7-2k$

$\Delta=0$

$b^2-4ac=0$

$(-2-k)^2-4(1)(-7-2k)=0$

$4+4k+k^2-4(-7-2k)=0$

$4+4k+k^2+28+8k=0$

$k^2+12k+32=0$

k 8 / k 4

$(k+8)(k+4)=0$

$k=-4,-8$

(7) $y=\frac{3x+1}{2x-4}$ $2x-4\neq 0$, $2x\neq 4$, $x\neq 2$

令 $t=2x-4$

则 $\frac{t+4}{2}=x$

$y=\frac{3(\frac{t+4}{2})+1}{t}$

$y=\frac{\frac{3t+12}{2}+1}{t}$

$y=\frac{\frac{3t}{2}+6+1}{t}$

$y=\frac{\frac{3t}{2}+7}{t}$

$y=\frac{3t}{2t}+\frac{7}{t}$

$y=\frac{3}{2}+\frac{7}{2x-4}$

$y=\frac{3}{2}+\frac{7}{2(x-2)}$

$y=\frac{3}{2}+\frac{7/2}{x-2}$

定义域: $\mathbb{R}\setminus\{2\}$

值域: $\mathbb{R}\setminus\{\frac{3}{2}\}$

单调性区间: $(-\infty,2)\cup(2,+\infty)$ ↘ 和

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(9) $y=x+m$ 与 $y=x^2$ 交点个数

$\begin{cases} y=x^2 \\ y=x+m \end{cases}$

$x^2=x+m$

$x^2-x-m=0$

无交点：$\Delta<0$

$(-1)^2-4(1)(-m)<0$

$1+4m<0$

$4m<-1$

$m<-\frac{1}{4}$

1个交点：$\Delta=0$

$(-1)^2-4(1)(-m)=0$

$1+4m=0$

$4m=-1$

$m=-\frac{1}{4}$

2个交点：$\Delta>0$

$(-1)^2-4(1)(-m)>0$

$1+4m>0$

$4m>-1$

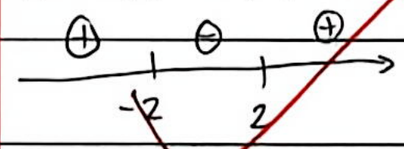
$m>-\frac{1}{4}$

(10) $f(x)=\sqrt{x}$，$g(x)=x^2-4$，$f(g(x))$?

$f(g(x))=\sqrt{x^2-4}$

定义域：$x^2-4\geq 0$

$(x-2)(x+2)\geq 0$



$x\in(-\infty,-2]\cup[2,+\infty)$

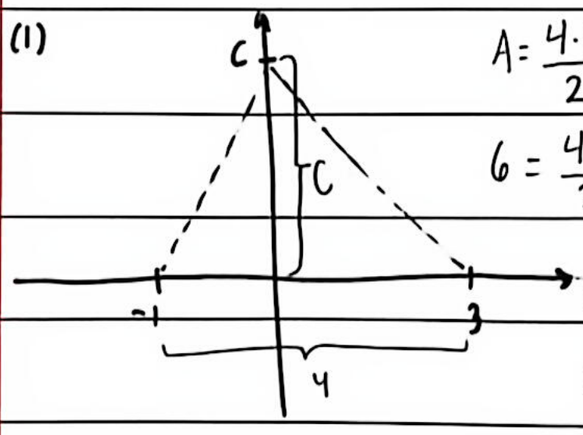
(11) $y=f(x)$

$f(x)=f(-x)$

$f(x)=f(-x)-3$

P.120 C组 1 2 5 6 7

(1)



$A=\frac{4\cdot C}{2}$ $6=2C$ $C=3$

$6=\frac{4C}{2}$ $3=C$ $(0,3)$

$a(x+1)(x-3)=y$

$a(0+1)(0-3)=3$

$a(1)(-3)=3$

$-3a=3$

$a=-1$ ——— a也可以是1

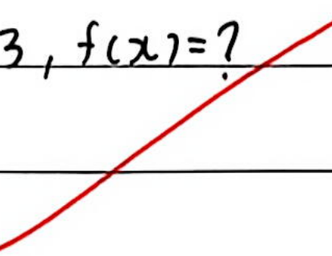
$f(x)=-(x+1)(x-3)$

(2) $y=mx+1$ 与 $y=\frac{x+1}{x}$ 只有一个交

题目是 $y=x+\frac{1}{x}$

$x\neq 0$

$y=1+\frac{1}{x}$



$\begin{cases} y=mx+1 \\ y=\frac{x+1}{x} \end{cases}$

$mx+1=\frac{x+1}{x}$

$mx^2+x=x+1$

$mx^2-1=0$ $m\neq 0$

其实m可以等于0

$\Delta=0$

$b^2-4ac=0$

$(0)^2-4(m)(-1)=0$

$0+4m=0$

$m=0$

无解，不能恰有一个公共点

(5) $f(x+1)=2x+3$，$f(x)=?$

$f(x)=2(x-1)+3$

$=2x-2+3$

$=2x+1$

(6) $f(x)=\frac{x-1}{x+1}$，求 $f(f(x))$ 和它的定义域

$f(f(x))=\frac{(\frac{x-1}{x+1})-1}{(\frac{x-1}{x+1})+1}=\frac{\frac{x-1}{x+1}-\frac{x+1}{x+1}}{\frac{x-1}{x+1}+\frac{x+1}{x+1}}=\frac{\frac{x-1-(x+1)}{x+1}}{\frac{x-1+x+1}{x+1}}=\frac{\frac{-2}{x+1}}{\frac{2x}{x+1}}$

$=\frac{-2}{x+1}\cdot\frac{x+1}{2x}=\frac{-2}{2x}=-\frac{1}{x}$，$(x\neq 0)$

$f(x)=\frac{x-1}{x+1}$，$(x\neq -1)$

所以 $f(f(x))=-\frac{1}{x}$ $(x\neq -1,0)$

(7) 若函数 f 又是奇又是偶，要满足什么条件（在定义域）

$f(x)=f(-x)=-f(-x)$

$f(-x)=0$，对所有x成立

$f(x)=0$，对所有x成立

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2026年10月23日

Page 118 A (4)(5)(7)(8)(9)(11)(14), Page 119 B (3)(7)(9)(10)(11), Page 120 C (1)(2)(5)(6)(7)

Page 118 A (4)(5)(7)(8)(9)(11)(14)

(4). Find the monotonic intervals (↳ vertex) and range (↳ vertex) of $y = \frac{-2}{a}x^2 + \frac{8}{b}x - \frac{5}{c}$

→ Vertex

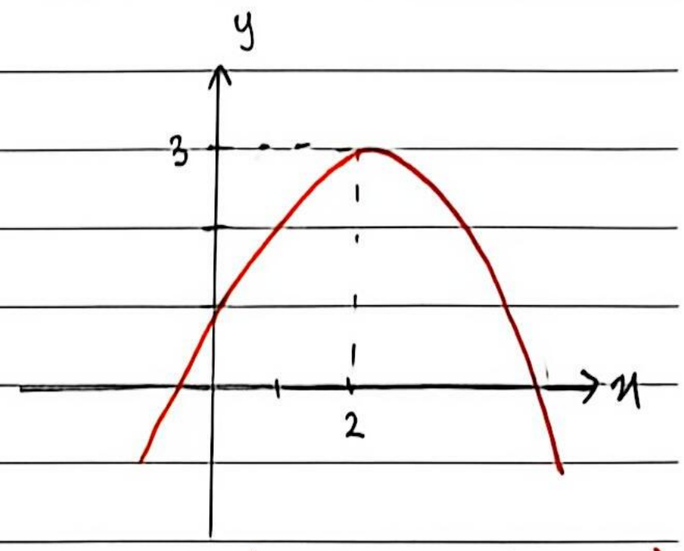
$x_p = \frac{-b}{2a} = \frac{-8}{2(-2)} = \frac{-8}{-4} = 2$ //

$y_p = \frac{b^2 - 4ac}{-4a} = \frac{8^2 - 4(-2)(-5)}{-4(-2)} = \frac{64 - 40}{8}$

$= \frac{24}{8}$

$= 3$ //

• Vertex : (2, 3)



up $(-\infty, 2)$ down $(2, +\infty)$ //

Range $(-\infty, 3]$ //

]

(5). Find the intersections of $y = x^2 - 2x - 3$ with the two coordinate axes

→ ◇ x intercept, $y = 0$

$0 = x^2 - 2x - 3$

$0 = (x + 1)(x - 3)$

$x + 1 = 0$ ∧ $x - 3 = 0$

$x = -1$ $x = 3$

Intersections $(-1, 0)$ 和 $(3, 0)$ //

◇ y intercept, $x = 0$

$y = 0^2 - 2(0) - 3$

$y = -3$

Intersections $(0, -3)$ //

(7). Write the domain, range, parity and monotonicity of $y = x^5$

→ Domain = $(-\infty, +\infty)$

Range = $(-\infty, +\infty)$

Parity = odd function

↳ $f(x) = -f(-x)$

$-f(-x) = -(-x)^5$

$-f(-x) = -(x^5)$

$-f(-x) = x^5$ //

$f(x) = -f(-x)$

$f(x) = x^5$

$f'(x) = 5x^4$

↳ always (+)

It is going up when $f'(x) > 0$

Increase on $(-\infty, +\infty)$ //

(8). Write the domain, range, and parity of $y = x^{-2}$ → $a^{-n} = \frac{1}{a^n}$

→ $y = x^{-2}$

$y = \frac{1}{x^2}$

◇ Domain

$x^2 \neq 0$

$x \neq 0$

Domain : $\{x | x \neq 0, x \in \mathbb{R}\}$

Domain : $(-\infty, 0) \cup (0, +\infty)$

◇ Range

$(0, +\infty)$

even function :

$f(x) = f(-x)$

$f(-x) = (-x)^{-2}$

$f(-x) = \frac{1}{(-x)^2}$

$f(-x) = \frac{1}{x^2} = x^{-2}$ //

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(10). $y = k/x$ passes through $(-3, 4)$. Find its analytic expression

→ $(-3, 4)$

x y

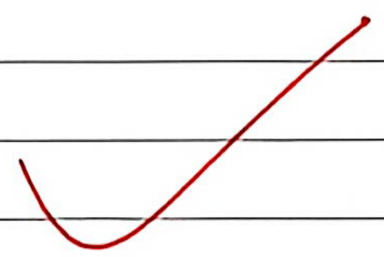
$y = \frac{k}{x}$

$4 = \frac{k}{-3}$

$k = -12$,

→ analytic expression $y = \frac{k}{x}$

$y = \frac{-12}{x}$ //



(11). Find the domain of $y = \frac{x^2 - 9}{x - 3}$ and explain the hole in the graph

→ $x - 3 \neq 0$

$x \neq 3$ //

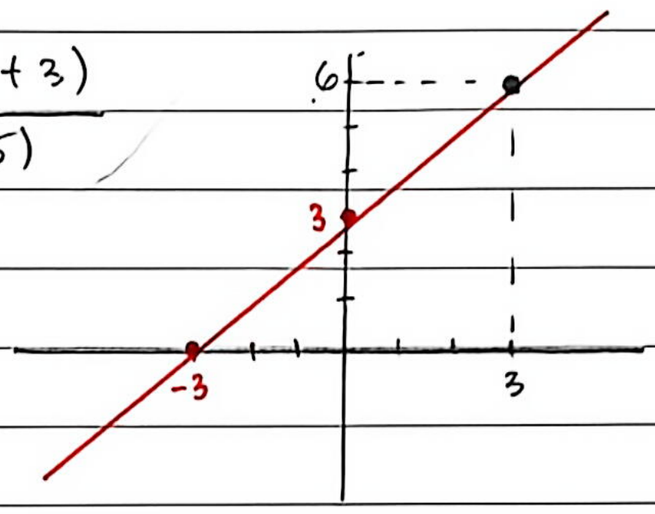
Domain $= (-\infty, 3) \cup (3, +\infty)$ //

$y = \frac{(x-3)(x+3)}{(x-3)}$

$y = x + 3$

$y = 3 + 3$

$y = 6$ //



(14). Shift $y = |x|$ left by 2 and up by 1

→

left by 3	right by 4	$(x, y) \xrightarrow{T\begin{bmatrix} a \\ b \end{bmatrix}} (x+a, y+b)$
$x + 3$	$x - 4$	
up by 4	down by 3	$x' = x + a$ $\quad y' = y + b$
$y - 4$	$y + 3$	$x' - a = x$ $\quad y' - b = y$

NOTE

NOTE

↳ $y = |x|$ left by 2, up by 1

$y - 1 = |x + 2|$ ↘ $x + 2$ ↳ $y - 1$

$y = |x + 2| + 1$ //

Page 119 ⅁ (3)(7)(9)(10)(11)

(3) The line $y = kx + 2k + 7$ and the parabola $y = x^2 - 2x$ have exactly one intersection. Find k.

→ $x^2 - 2x = kx + 2k + 7$

$x^2 - (k+2)x - (2k+7) = 0$

$a = 1$, $b = -(k+2)$, $c = -(2k+7)$

$\Delta = b^2 - 4ac = 0$

$= (-(k+2))^2 - 4(1)(-(2k+7)) = 0$

$= (k+2)^2 + 8k + 28 = 0$

$= k^2 + 4k + 4 + 8k + 28 = 0$

$k^2 + 12k + 32 = 0$

factorise:

$(k+4)(k+8) = 0$

$k + 4 = 0$ 或 $k + 8 = 0$

$k = -4$ 或 $k = -8$

$k = -4$ 或 $k = -8$ //

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119 B (5)(7)(9)(10)(11)

). Find the domain, range, asymptotes and monotonic intervals of $y = \frac{3x+1}{2x-4}$

→ Domain : $2x - 4 \neq 0$

$2x \neq 4$

$x \neq 2$ //

$\mathbb{R} \setminus \{2\}$

$= (-\infty, 2) \cup (2, +\infty)$ //

Range : $y(2x-4) = 3x+1$

$2xy - 4y = 3x + 1$

$2xy - 3x = 4y + 1$

$x(2y-3) = 4y + 1$

$\mathbb{R} \setminus \{\frac{3}{2}\}$

$x = \frac{4y+1}{2y-3}$

$2y - 3 \neq 0$

$2y \neq 3$

$y \neq \frac{3}{2}$ //

Range : $(-\infty, \frac{3}{2}) \cup (\frac{3}{2}, +\infty)$ //

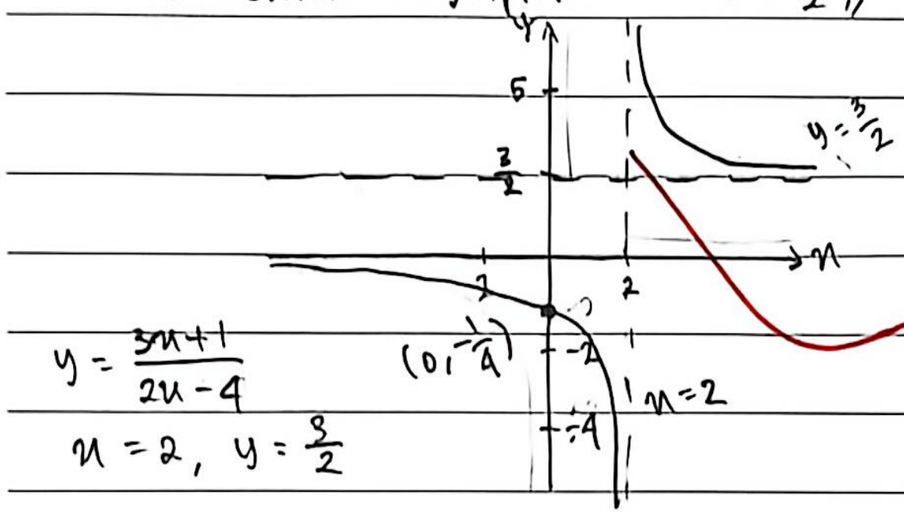
Asymptotes

• vertical Asymptote = $2x - 4 = 0$

$2x = 4$

$x = 2$ //

• Horizontal Asymptote $y = \frac{3}{2}$ //



Monotonic Intervals → use quotient rule

$y' = \frac{uv' - vu'}{v^2}$

→ $y' = \frac{3(2x-4) - 2(3x+1)}{(2x-4)^2}$

$y' = \frac{6x - 12 - 6x - 2}{(2x-4)^2}$

$y' = \frac{-14}{(2x-4)^2}$

$y' < 0$ for all $x \neq 2$

the function is decreasing on both intervals

$(-\infty, 2)$ 和 $(2, +\infty)$ //

(9). Discuss the number of intersections between the line $y = x + m$ and the parabola $y = x^2$

→ $x + m = x^2$

$\underset{a}{x^2} - \underset{b}{x} - \underset{c}{m} = 0$

$\Delta = b^2 - 4ac$

$= (-1)^2 - 4(1)(-m)$

$= 1 + 4m$ //

•) $\Delta > 0$ → two intersections

$1 + 4m > 0$ → $m > -\frac{1}{4}$ //

•) $\Delta = 0$ → one intersection

$1 + 4m = 0$ → $m = -\frac{1}{4}$ //

•) $\Delta < 0$ → no real intersections

$1 + 4m < 0$ → $m < -\frac{1}{4}$ //

(10) $f(x) = \sqrt{x}$, $g(x) = x^2 - 4$. Find the domain of $f(g(x))$.

→ $f(g(x)) = \sqrt{x^2 - 4}$

$x^2 - 4 \geq 0$

$x^2 \geq 4$

$x \leq -2$ 或 $x \geq 2$ // $x \geq \pm 2$ // → $(-\infty, -2] \cup [2, +\infty)$ //

(11). Given the graph of $y = f(x)$, reflect it about the y-axis first, then shift it down by 3. Write the new function.

→ replace x with $-x$

$y = f(-x)$

$y = f(-x) - 3$ //

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(1). A Quadratic function intersects the x-axis at $(-1, 0)$ and $(3, 0)$, and intersects the y-axis at C. If the area of the triangle is 6, find the equation of the parabola.

→ the roots are $x = -1$ and $x = 3$

$y = a(x+1)(x-3)$

• intersection with y axis, $x = 0$

$y = a(0+1)(0-3) =$

$y = a(-3)$

$y = -3a$ //

↳ $C = (0, -3a)$ //

$A = (-1, 0)$, $B = (3, 0)$

$AB = 3 - (-1) = 4$ //

$A = \frac{1}{2} \times b \times \text{height}$

$6 = \frac{1}{2} \times 4 \times \text{height}$

$6 = 2 \times \text{height}$

$\text{height} = \frac{6}{2} = 3$ //

$h = |-3a| = 3|a|$

$h = 3$

$\rightarrow \frac{1}{2} \times 4 \times 3|a| = 6$

$6|a| = 6$

$|a| = 1$ //

so, $a = 1$ 或 $a = -1$ //

case 1 : $a = 1$ → $y = (x+1)(x-3)$

$y = x^2 - 2x - 3$ //

case 2 : $a = -1$ → $y = -(x+1)(x-3)$

$y = -x^2 + 2x + 3$ //

There are two possible equations : $y = x^2 - 2x - 3$ 或 $y = -x^2 + 2x + 3$ //

(2). Find all m such that the line $y = mx + 1$ and $y = x + 1/x$ have exactly one common point.

→ $mx + 1 = x + \frac{1}{x}$ $(x \neq 0)$ ~ so multiply both sides by x

$= mx^2 + x = x^2 + 1$

$= \underset{a}{(m-1)}x^2 + \underset{b}{x} \underset{c}{-1} = 0$

case 1 : $m \neq 1$

$\Delta = b^2 - 4ac$

$= 1^2 - 4(m-1)(-1) = 0$

$= 1 + 4(m-1) = 0$

$= 1 + 4m - 4 = 0$ | $4(m-1) = 4m - 4$

$= 4m + 1 - 4 = 0$ ~ $1 - 4 = -3$

↳ $4m - 3 = 0$

$4m = 3$

$m = \frac{3}{4}$ //

case 2 : $m = 1$

$x - 1 = 0$

$x = 1$ //

$m = \frac{3}{4}$ 或 $m = 1$ //

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). Given $f(x+1) = 2x + 3$, And $f(x)$

→ change $x+1$ into t

$x + 1 = t$

$x = t - 1$

Substitute $x = t - 1$ into the equation

$f(x+1) = 2x + 3$

↳ $f(t) = 2(t-1) + 3$

$f(t) = 2t - 2 + 3$

$f(t) = 2t + 1$

Replace t with x → $f(x) = 2x + 1$ //

(6). Let $f(x) = \frac{x-1}{x+1}$. find $f(f(x))$, and write the domain restrictions completely

→ $f(f(x)) = \frac{f(x) - 1}{f(x) + 1}$

Substitute $f(x) = \frac{x-1}{x+1}$ → $f(f(x)) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1}$

Simplify the numerator

$\frac{x-1}{x+1} - 1 = \frac{x - 1 - (x+1)}{x+1}$

$= \frac{x - 1 - x - 1}{x+1} = \frac{-2}{x+1}$

$\frac{x-1}{x+1} + 1 = \frac{x - 1 + (x+1)}{x+1}$

$= \frac{x - 1 + x + 1}{x+1} = \frac{2x}{x+1}$

$f(f(x)) = \frac{\frac{-2}{x+1}}{\frac{2x}{x+1}} = \frac{-2}{x+1} \times \frac{x+1}{2x}$

cancel the common factors : $f(f(x)) = -\frac{1}{x}$

$x + 1 \neq 0$ -	$f(x) \neq -1$
$x \neq -1$ //	$\frac{x-1}{x+1} \neq -1$

Since $x \neq -1$, multiply by $x + 1$

$x - 1 \neq -(x+1)$

$x - 1 \neq -x - 1$

$2x \neq 0 \rightarrow x \neq 0$

the complete domain restrictions are $x \neq -1$, $x \neq 0$ //

interval notation = $(-\infty, -1) \cup (-1, 0) \cup (0, +\infty)$ //

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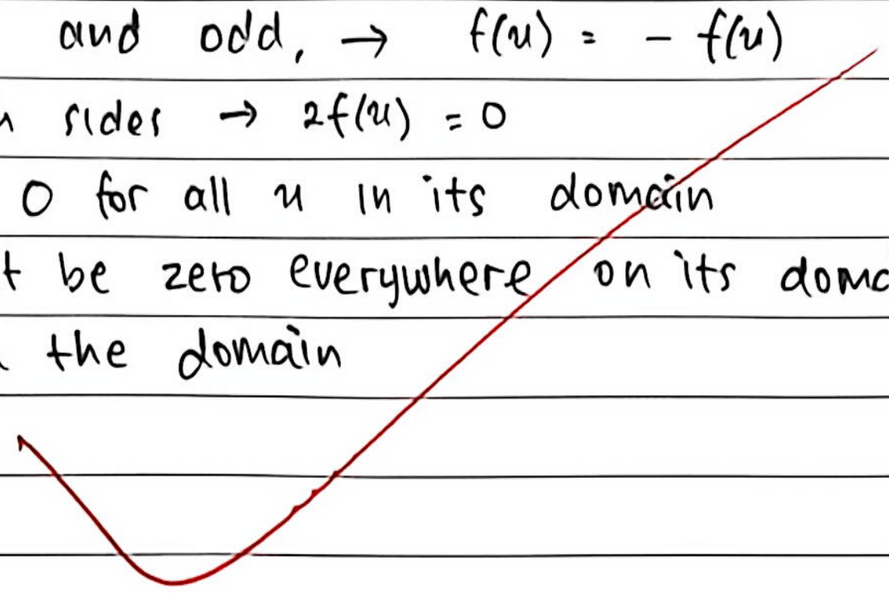
(7). If a function f is both odd and even, explain what condition f(x) must satisfy throughout its domain

→ f is both even and odd, → $f(x) = -f(x)$

add f(x) to both sides → $2f(x) = 0$

therefore, $f(x) = 0$ for all x in its domain

↝ the function must be zero everywhere on its domain, → $f(x) = 0$ for every x in the domain



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